University of Regensburg Institute of Theoretical Physics

Boundary-Induced Orbital Hall Effect from Intersite Currents in Finite Honeycomb Lattices

Research Project Report

by

Leonard Prokisch

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Supervisors: Nithin Thomas

Prof. Dr. Ferdinand Evers

Abstract

Monolayer graphene is known to not exhibit any bulk orbital Hall effect (OHE). However, ab initio simulations of armchair nanoribbons make it plausible that open boundaries can induce a finite OHE: stronger current streamlines at the edges lead to plaquettes with a net ring current component. To study the relevance of their orbital angular momentum (OAM), we evaluate the OAM operator in a tight-binding model using intersite currents and apply the muffin-tin truncation from the atom-centered approximation on the plaquette level. We arrive at a conceptually simple real space method to estimate the orbital Hall conductivity (OHC) of finite graphene-like systems. We find that the current streamlines indeed induce a relevant OHE with OHC similar to periodic gapped graphene that increases further for resonant high-frequency driving.

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1 Introduction

Data transfer and storage are central achievements of our digital society. It is fascinating how past research enabled us to understand the movement of electrons so precisely that we can use charge to store information and electric currents to transport it. To improve the efficiency and stability of the according devices, scientists use the electron's spin as an additional degree of freedom in the field of spintronics [1]. The foundation is usually the conversion between spin and orbital angular momentum (OAM) from charge currents via spin orbit interaction [2].

It seems natural to also investigate the OAM degree of freedom of electrons. This field has indeed emerged and was named *orbitronics* [3]. Experimental evidence for orbital currents was found recently by a magneto-optical measurement of the orbital Hall effect (OHE) [4]. The OHE is central for orbitronics as it describes the generation of an orbital current by a transverse electric field [5].

The boundaries of finite systems such as armchair graphene nanoribbons (AGNRs) can contribute to ring currents that carry OAM [6, 7]. Thus, it seems plausible that open boundary conditions (OBC) can introduce an OHE in materials with otherwise vanishing orbital Hall conductivity (OHC). In this research project, we therefore investigate whether open boundaries induce a nonzero OHC in a finite graphene-like system driven by an alternating electric field. To calculate the orbital moment in real space, we follow the idea of muffin-tin truncation from the atom-centered approximation (ACA) that is normally employed for intrasite contributions [8] and apply it to intersite currents. This allows a conceptually simple estimate of the OHC to find out if a boundary-induced OHE is relevant.

This report is structured as follows. Section 2 explains how existing simulations hint at a possible boundary-induced OHE in AGNRs and why it is non-trivial to capture with usual orbitronics methods. Next, section 3 sets up the tight-binding model of a finite honeycomb lattice on which we evaluate the OAM operator in section 4 using intersite currents around single plaquettes. Section 5 then describes how the time evolution of the systems is obtained numerically. Finally, section 6 presents the obtained OHCs from the simulated systems and investigates the connection with the bond current patterns for different system dimensions and driving frequencies. This allows us to draw a conclusion in section 7 on the relevance of an intersite current OHE from OBC.

2 Motivation

2.1 Boundary Effects in Graphene Nanoribbons

While the OHE vanishes in periodic graphene [9], the open boundaries of an AGNR lead to an interesting phenomenon: the current density in fig. 1 obtained via *ab initio* density functional theory simulations is, as one might expect intuitively, higher at the open armchair boundaries. It follows that currents around the hexagonal plaquettes at the edges have a net ring current component. Their sense of rotation is opposite for the two armchair edges and would invert if we reverse the voltage. Hence, components perpendicular to the ribbon plane of the OAM from the plaquette ring currents have opposite sign at the

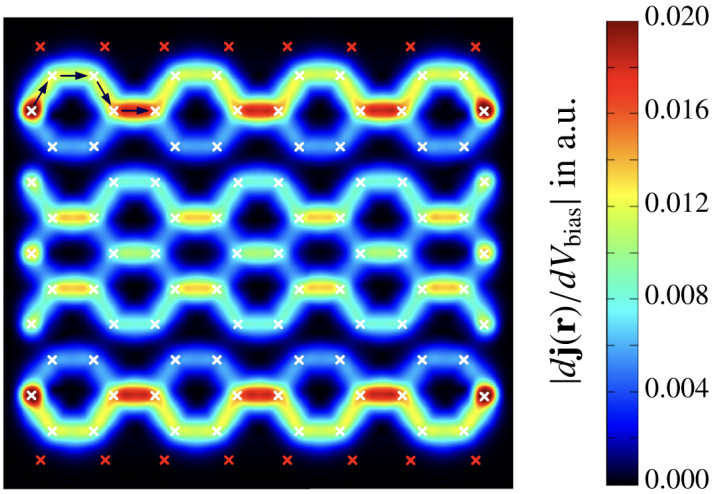

Figure 1. Transport calculation of the intensity of the local current density in an AGNR. The colormap shows the current magnitude in the plane 0.05 nm above the ribbon. The arrows indicate the flow direction. At the boundary, streamlines of stronger currents are visible. Figure taken from Ref. [6].

two edges and flip when the voltage is reversed. This fits the situation of an OHE, where OAM is transported perpendicular to an applied electric field via an orbital current.

2.2 What is an Orbital Current and how is it calculated?

The orbital current operator measures the flow of OAM and is defined as

$$\hat{\boldsymbol{\jmath}}^{\gamma}_{\text{orb}} := \hat{\mathbf{v}}\,\hat{L}^{\gamma} + \hat{L}^{\gamma}\,\hat{\mathbf{v}} \tag{1}$$

with the velocity operator $\hat{\mathbf{v}}$ and the γ-component of the OAM operator $\hat{\mathbf{L}}$ [5]. A practical example of an orbital current is a wheel rolling through our coordinate system with constant velocity [9]: in coordinates where the origin is shifted to the wheel's center of mass, its OAM is constant. From our origin, its OAM varies with the change corresponding to an orbital current. This example hints at how to understand eq. (1), the orbital current density at position $\mathbf{r}$ is determined by the OAM with respect to $\mathbf{r}$ as the origin.

In tight-binding models, orbital currents are often calculated in the ACA. $\hat{\boldsymbol{\jmath}}^{\gamma}_{\text{orb}}$ is defined at the discrete set of atom site locations $(\mathbf{r}_i)$ where $\hat{\mathbf{L}}$ is evaluated as the local OAM operator of the isolated atom, $\hat{\mathbf{L}}(\mathbf{r}_i) := m_e\,(\hat{\mathbf{r}} - \mathbf{r}_i) \times \hat{\mathbf{v}}$ with the electron mass m_e and the position operator $\hat{\mathbf{r}}$ [10]. The contributions from other atom sites are neglected, i.e., $\hat{\mathbf{L}}$ is calculated within the atom spheres or "muffin tins" [8]. A consequence of this definition is that the ACA requires orbitals with different $\hat{L}^{\gamma}$ eigenvalues for a finite orbital current.

One can include nonlocal contributions using the *modern theory of orbital magnetization*. The OAM operator is defined in terms of the orbital magnetic moment $\hat{\mathbf{m}}$, $\hat{\mathbf{L}} := -2\,m_e\,\hat{\mathbf{m}}$, with $\hat{\mathbf{m}} = -\frac{e}{4}(\hat{\mathbf{r}} \times \hat{\mathbf{v}} - \hat{\mathbf{v}} \times \hat{\mathbf{r}})$ and the elementary charge e; the proportionality uses the Landé g-factor $g_L = 1$ [11]. The magnetic moment is calculated in momentum space via the orbital Berry curvature [10, 12].

We avoid using the modern theory in real space as it is involved [13] and outside this project's scope. The ACA has also been used for real space calculations [14] but is not applicable in our graphene-like system where all atom sites have identical orbital character.

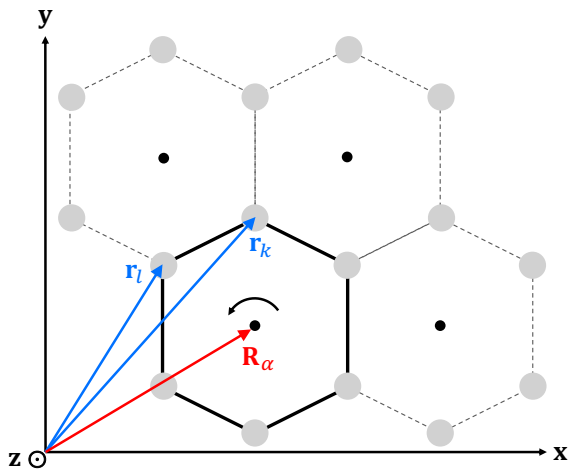

Figure 2. The used coordinate system for the honeycomb lattice in the xy-plane. The grey circles denote the atom site positions with coordinates ($\mathbf{r}_i$). The black dots are the plaquette centers at the Bravais lattice points ($\mathbf{R}_\alpha$). The solidly outlined black hexagon is a plaquette. Counter-clockwise sums over bonds as in eq. (16) are meant with respect to plaquette centers, indicated by the curved arrow for the shown plaquette $\mathbf{R}_\alpha$, e.g., the bond $\mathbf{r}_l - \mathbf{r}_k$ points in the stipulated counter-clockwise direction.

Instead, we use the muffin-tin idea from the ACA and apply it to intersite ring currents. This means we evaluate $\hat{\mathbf{L}}$ using the particle current operator but truncate contributions to a single plaquette. We neglect larger-scale ring currents and assume that currents on the local plaquette are the relevant scale for an OHE induced by OBC.

3 Model

In order to calculate the orbital current in the just described way, we set up a tight-binding model for a finite graphene-like honeycomb lattice in the following. This structure is suitable since it does not exhibit an OHE in the ACA or, for the periodic case, in the modern theory [9]. We can thus attribute a potential OHE to the OBC and the intersite currents.

3.1 Lattice Geometry

The honeycomb structure is a hexagonal Bravais lattice with a two-atom basis. We choose coordinates where the Bravais lattice points (at ($\mathbf{R}_\alpha$)) lie at the center of a hexagonal plaquette formed by six atoms (at ($\mathbf{r}_i$)). The points and coordinate axis are illustrated in fig. 2. Boundaries are open along both the x- and y-direction, resulting in armchair edges parallel to the y-axis. System dimensions are denoted by the number of dimer lines L_x and the number of zigzag rows L_y. The distance of nearest neighbor atoms is set to $a_{\mathrm{NN}} = 1$ and the atom sites are indexed from 1 to $N := L_x\, L_y$.

A site basis vector $|i\rangle$ denotes the state where only the i-th atom site is occupied by an electron. In general, we do not break any sublattice symmetry and assume uniform on-site energy: the states ($|i\rangle$) are translations of the same isolated atom state for which we assume an s-like orbital.

3.2 Tight-Binding Hamiltonian

We assume electrons to be strongly localized at single atom sites, such that

- $(|i\rangle)$ is an orthonormal basis of our Hilbert space,
- the position operator $\hat{\mathbf{x}}$ is diagonal in the site basis, $\langle i|\,\hat{\mathbf{x}}\,|j\rangle = \delta_{ij}\mathbf{r}_i$,
- matrix elements $\langle i|\,\hat{\mathcal{H}}_0\,|j\rangle$ of the unperturbed Hamiltonian $\hat{\mathcal{H}}_0$ are zero if the sites i, j are of greater spatial distance than nearest neighbors.

We set the diagonal part of $\hat{\mathcal{H}}_0$ to zero and assume that the nearest-neighbor hopping energy t_{hop} is uniform for all nearest-neighbor bonds. The unperturbed hopping Hamiltonian $\hat{\mathcal{H}}_0$ reads

$$\hat{\mathcal{H}}_0 = -t_{\text{hop}} \sum_{\langle i,j\rangle} |i\rangle\langle j| + \text{h.c.} \tag{2}$$

where $\langle i, j\rangle$ denotes the set of all bonds between nearest neighbor sites i, j.

We apply a time dependent homogeneous electric field $\mathbf{E}(t)$ parallel to the y-axis with magnitude $\epsilon(t)$,

$$\mathbf{E}(\mathbf{r}, t) = \epsilon(t)\mathbf{y} = -\nabla\epsilon(t)(-\mathbf{y}\cdot\mathbf{r}). \tag{3}$$

If we set the electron charge in the potential to $-e = -1$, we arrive at the time-dependent total Hamiltonian $\hat{\mathcal{H}}(t)$ via onsite potentials:

$$\hat{\mathcal{H}}(t) = \hat{\mathcal{H}}_0 + \hat{\mathcal{H}}_{\text{ext}}(t), \quad \hat{\mathcal{H}}_{\text{ext}}(t) := \epsilon(t) \sum_i \mathbf{y}\cdot\mathbf{r}_i\ |i\rangle\langle i| \tag{4}$$

$\hat{\mathcal{H}}_{\text{ext}}(t)$ models an AC field with amplitude E_0 and angular frequency ω. The field is switched on slowly over a time T_{ramp} as

$$\epsilon(t) = E_0 \sin(\omega t) \cdot \begin{cases} 0 & \text{for } t < 0 \\ \sin^2\left(\frac{\pi t}{2\,T_{\text{ramp}}}\right) & \text{for } 0 \le t < T_{\text{ramp}} \\ 1 & \text{for } T_{\text{ramp}} \le t \end{cases} \tag{5}$$

3.3 Electronic Properties

In our finite system of N atoms we expect N energy eigenstates. Most of these turn out as states that are superpositions of Bloch states with opposite crystal momenta $\pm k_x, \pm k_y$ [15, 16] such that $k_{x,y}$ are still useful labels despite the violation of Bloch's theorem. Their energies are described well by the dispersion of the infinite crystal [16] (for which we expect two energy bands from the two-atom basis). Since we have both open armchair and zigzag boundaries, we expect an energy gap for the bulk states that contains states of energy $E \approx 0$, such as localized edge states [15]. Solving the eigenvalue problem for our Hamiltonian (2) confirms this: fig. 3 shows the spectrum and two eigenstates of $\hat{\mathcal{H}}_0$ for an example system. Indeed, the eigenenergies jump across $E = 0$, with two $E \approx 0$ edge states where the density is localized at the zigzag boundaries. We can also see a large number of states with $E = \pm t_{\text{hop}}$, these stem from the respective van Hove singularities in the density of states [17].

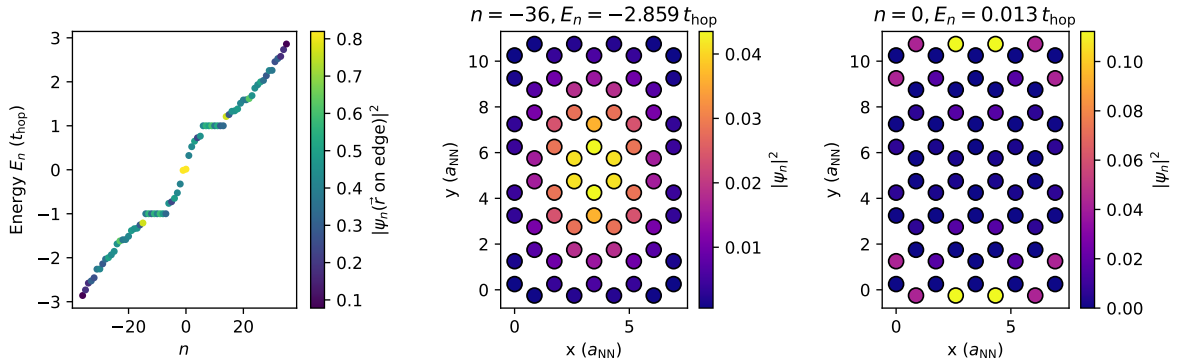

Figure 3. Spectrum and two eigenstates of the hopping Hamiltonian $\hat{\mathcal{H}}_0$ for a system with $(L_x, L_y) = (9, 8)$. The dots in the left plot are the eigenenergies (E_n) sorted from smallest to largest, the dot colormap indicates how much the according eigenstate $|\psi_n\rangle$ is concentrated on the lattice edge (sum of the boundary site densities $\sum_{i \text{ is on edge}} |\langle i|\psi_n\rangle|^2$). The center plot shows the real space densities of the lowest energy state, the right plot analogously shows an $E \approx 0$ state. Sites are plotted as circles at their real space x, y-coordinates, the colormap corresponds to the respective density $|\langle i|\psi_n\rangle|^2$.

4 Orbital Angular Momentum Operator

We now define the local OAM operator $\hat{\boldsymbol{\ell}}(\mathbf{r})$ for the tight-binding model. As described in section 2.2, we start with the usual OAM operator $\hat{\mathbf{L}}(\mathbf{r}_0) = m_e\,(\hat{\mathbf{r}} - \mathbf{r}_0) \times \hat{\mathbf{v}}$ but truncate contributions to intersite currents on the local plaquette. Intraatomic contributions vanish since we assume identical s-like orbitals at each atom site.

4.1 Probability Current

First, we rewrite $\hat{\mathbf{L}} := \hat{\mathbf{L}}(\mathbf{0}) = m_e\,\hat{\mathbf{r}} \times \hat{\mathbf{v}}$ using the probability current. Denoting the current from site l to site k by $\hat{J}_{k\leftarrow l}$ and the particle density at site k by $\hat{n}_k := |k\rangle\langle k|$, the continuity equation

$$\frac{\mathrm{d}\hat{n}_k}{\mathrm{d}t} = \sum_{l\neq k} \hat{J}_{k\leftarrow l}. \tag{6}$$

leads to the operator equation

$$\hat{J}_{k\leftarrow l} = i(\mathcal{H}_{lk}\,|l\rangle\langle k| - \mathcal{H}_{kl}\,|k\rangle\langle l|), \tag{7}$$

where matrix elements $\mathcal{H}_{kl} := \langle k|\,\hat{\mathcal{H}}\,|l\rangle$ were abbreviated. We can read off the current operator matrix elements:

$$(J_{k\leftarrow l})_{ij} = i\,\mathcal{H}_{ij}\delta_{li}\delta_{kj} - i\,\mathcal{H}_{ij}\delta_{ki}\delta_{lj}. \tag{8}$$

Also, from the Heisenberg equation of motion and the diagonality of the position operator (assumed in section 3.2), one obtains the velocity operator matrix elements as

$$\mathbf{v}_{ij} = i\,[\hat{\mathcal{H}}, \hat{\mathbf{r}}]_{ij} = i\,(\mathbf{r}_j - \mathbf{r}_i)\,\mathcal{H}_{ij}, \tag{9}$$

which provides the OAM operator matrix elements written using the probability current:

$$\mathbf{L}_{ij} = m_e\,\mathbf{r}_i \times \mathbf{v}_{ij} = i m_e\,(\mathbf{r}_i \times \mathbf{r}_j)\mathcal{H}_{ij} = -m_e \sum_{k<l} (\mathbf{r}_k \times \mathbf{r}_l)\,(J_{k\leftarrow l})_{ij}, \tag{10}$$

where the sum $\sum_{k<l}$ runs over all unique pairs of atom sites in our system. Since our Hamiltonian (eq. (4)) only has non-zero off-diagonal matrix elements for nearest neighbor sites, we obtain the final result

$$\hat{\mathbf{L}} = -m_e \sum_{\langle k,l \rangle} (\mathbf{r}_k \times \mathbf{r}_l)\, \hat{J}_{k \leftarrow l}. \tag{11}$$

4.2 Orbital Angular Momentum of a Plaquette

We now obtain the local OAM density operator $\hat{\boldsymbol{\ell}}(\mathbf{r})$ from $\hat{\mathbf{L}}(\mathbf{r})$ by truncating all contributions outside the plaquette around $\mathbf{r}$. As introduced in section 3.1, the plaquettes are identified by the Bravais lattice points $\mathbf{R}_\alpha$ and have the shape of the smallest regular hexagon whose vertices lie on atom sites $\mathbf{r}_i$. Denoting the set of the six nearest-neighbor pairs that form the edges of the plaquette whose center is closest to $\mathbf{r}$ by $\circlearrowleft_{\mathbf{r}}$, we have

$$\hat{\boldsymbol{\ell}}(\mathbf{r}) = -\frac{m_e}{A_{\circlearrowleft}} \sum_{(k,l) \in \circlearrowleft_{\mathbf{r}}} (\mathbf{r}_k - \mathbf{r}) \times (\mathbf{r}_l - \mathbf{r})\, \hat{J}_{k \leftarrow l}, \tag{12}$$

where we normalized by the plaquette area $A_{\circlearrowleft}$.

It is insightful to evaluate $\hat{\boldsymbol{\ell}}(\mathbf{r})$ at the plaquette level, i.e., for the $(\mathbf{R}_\alpha)$. Denoting the integral over a plaquette hexagon with center at the origin by $\int_{A_{\circlearrowleft}} \mathrm{d}^2 s$, we calculate

$$\begin{aligned}
&\int_{A_{\circlearrowleft}} \mathrm{d}^2 s\, (\mathbf{r}_k - \mathbf{R}_\alpha - \mathbf{s}) \times (\mathbf{r}_l - \mathbf{R}_\alpha - \mathbf{s}) \\
&\quad = (\mathbf{r}_k - \mathbf{R}_\alpha) \times (\mathbf{r}_l - \mathbf{R}_\alpha) \int_{A_{\circlearrowleft}} \mathrm{d}^2 s + \int_{A_{\circlearrowleft}} \mathrm{d}^2 s\, (\mathbf{r}_l - \mathbf{r}_k) \times \mathbf{s} \\
&\quad = A_{\circlearrowleft}\, (\mathbf{r}_k - \mathbf{R}_\alpha) \times (\mathbf{r}_l - \mathbf{R}_\alpha) =: A_{\circlearrowleft}\, \theta_{kl}^{(\alpha)}.
\end{aligned} \tag{13}$$

With eq. (13), we obtain the OAM contribution of the plaquette at $\mathbf{R}_\alpha$:

$$\frac{1}{A_{\circlearrowleft}} \int_{A_{\circlearrowleft}} \mathrm{d}^2 s\, \hat{\boldsymbol{\ell}}(\mathbf{R}_\alpha + \mathbf{s}) = \hat{\boldsymbol{\ell}}(\mathbf{R}_\alpha) = -\frac{m_e}{A_{\circlearrowleft}} \sum_{(k,l) \in \circlearrowleft_{\mathbf{R}_\alpha}} \theta_{kl}^{(\alpha)}\, \hat{J}_{k \leftarrow l}. \tag{14}$$

Since $\mathbf{r}_k, \mathbf{r}_l$ are neighboring vertices of a regular hexagon with sidelength $a_{\mathrm{NN}} = 1$ in the xy-plane, we obtain

$$\theta_{kl}^{(\alpha)} = a_{\mathrm{NN}}^2 \sin \frac{2\pi}{6}\, \mathbf{z} = \frac{\sqrt{3}}{2}\, \mathbf{z}, \tag{15}$$

if the bond $\mathbf{r}_l - \mathbf{r}_k$ points counter-clockwise with respect to $\mathbf{R}_\alpha$ when looking against the z-axis, as is the case in fig. 2. Thus, after ordering the pairs $(k,l) \in \circlearrowleft_{\mathbf{R}}$ counter-clockwise in this way and plugging $A_{\circlearrowleft} = \frac{3\sqrt{3}}{2} a_{\mathrm{NN}}^2$ for the hexagon area, the average z-OAM density of a plaquette is given by

$$\hat{\ell}^{(z)}(\mathbf{R}_\alpha) = -\frac{m_e}{3} \sum_{(k,l) \in \circlearrowleft_{\mathbf{R}_\alpha}} \hat{J}_{k \leftarrow l}. \tag{16}$$

This expression indeed corresponds to the directed ring currents that we saw in the AGNR in section 2.1!

4.3 Orbital Polarization Current

We can now use the OAM density $\hat{\boldsymbol{\ell}}$ to look for an orbital current. From now on, we always use the z-components $\hat{\ell}^{(z)}, \hat{L}^{(z)}$ and $\hat{\boldsymbol{\jmath}}^{(z)}_{\text{orb}}$ since they are the only nontrivial ones for our 2D system.

For simplicity, we work with the global polarization current, which requires a continuity equation,

$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\ell}^{(z)}(\mathbf{r}) + \nabla \cdot \hat{\boldsymbol{\jmath}}^{(z)}_{\text{orb}}(\mathbf{r}) = 0. \tag{17}$$

Equation (17) is likely invalid in practice without source terms, since the hard truncation of contributions outside a single plaquette is somewhat arbitrary. Note however the similarity between $\hat{\boldsymbol{\ell}}(\mathbf{r})$ and the magnetic flux density $\mathbf{B}(\mathbf{r})$ in the Biot-Savart law,

$$\mathbf{B}(\mathbf{r}) \propto \int \mathrm{d}^3r' \, \frac{1}{|\mathbf{r}-\mathbf{r}'|^3} \, (\mathbf{r}-\mathbf{r}') \times \mathbf{j} \tag{18}$$

where the OAM of electrons $\propto (\mathbf{r}-\mathbf{r}') \times \mathbf{j}$ with the electric current density $\mathbf{j}$ and the origin $\mathbf{r}$ is integrated such that currents far from $\mathbf{r}$ have small weights. Interestingly, the time derivative of the flux density $\dot{\mathbf{B}}$ is a pure divergence (component-wise) as seen from the Maxwell-Faraday equation, $\dot{\mathbf{B}} + \nabla \times \boldsymbol{\varepsilon} = 0$ with the electric field $\boldsymbol{\varepsilon}$ [18]. However, the conservation of orbital currents is a matter of ongoing research [19, 20]. To move forward, we accept errors from finite source terms by assuming eq. (17) is valid and define the orbital polarization $\hat{\mathbf{P}}_{\text{orb}}$ as

$$\hat{\mathbf{P}}_{\text{orb}} := \frac{1}{A_{\text{tot}}} \int_{A_{\text{tot}}} \mathrm{d}^2r \, \hat{\ell}^{(z)}(\mathbf{r}) \cdot \mathbf{r}, \tag{19}$$

where A_{tot} is the total area of the honeycomb system. If the magnitude of the local OAM $\ell^{(z)}$ is indeed larger at the boundaries and polarized as expected in section 2.1, a simultaneous sign flip of $\ell^{(z)}$ over the whole lattice following the electric AC field results in a finite orbital polarization that also flips with the electric field. With eq. (17) we assume an orbital current as the sole cause and neglect potential other sources of OAM. Then the average orbital current of the whole system $\hat{\mathbf{J}}^{(z)}_{\text{orb}}$ is

$$\begin{aligned}\hat{\mathbf{J}}^{(z)}_{\text{orb}} &= \frac{1}{A_{\text{tot}}} \int_{A_{\text{tot}}} \mathrm{d}^2r \, \hat{\boldsymbol{\jmath}}^{(z)}_{\text{orb}}(\mathbf{r}) = -\frac{1}{A_{\text{tot}}} \int_{A_{\text{tot}}} \mathrm{d}^2r \, \mathbf{r} \, \nabla \cdot \hat{\boldsymbol{\jmath}}^{(z)}_{\text{orb}}(\mathbf{r}) = \frac{1}{A_{\text{tot}}} \int_{A_{\text{tot}}} \mathrm{d}^2r \, \mathbf{r} \, \frac{\mathrm{d}}{\mathrm{d}t}\hat{\ell}^{(z)}(\mathbf{r}) \\ &= \frac{\mathrm{d}}{\mathrm{d}t}\hat{\mathbf{P}}_{\text{orb}}.\end{aligned} \tag{20}$$

4.4 Aspects of the Orbital Polarization

Before we quantify the orbital current response, we look at two interesting decompositions of the orbital polarization in eq. (19): one can obtain a part that depends only on loop current sums like eq. (16) or a bulk term that is independent of the coordinate origin.

We start by evaluating the position dependent terms in the integral over a single plaquette $\mathbf{R}_\alpha$:

$$\begin{aligned}
&\int_{A_{\hexagon}} \mathrm{d}^2 s \left[(\mathbf{r}_k - \mathbf{R}_\alpha - \mathbf{s}) \times (\mathbf{r}_l - \mathbf{R}_\alpha - \mathbf{s})\right]^{(z)} (\mathbf{R}_\alpha + \mathbf{s}) \\
&\quad = \left[(\mathbf{r}_k - \mathbf{R}_\alpha) \times (\mathbf{r}_l - \mathbf{R}_\alpha)\right]^{(z)} \int_{A_{\hexagon}} \mathrm{d}^2 s\, (\mathbf{R}_\alpha + \mathbf{s}) + \int_{A_{\hexagon}} \mathrm{d}^2 s \left[(\mathbf{r}_l - \mathbf{r}_k) \times \mathbf{s}\right]^{(z)} (\mathbf{R}_\alpha + \mathbf{s}) \\
&\quad = A_{\hexagon} \left(\theta_{kl}^{(\alpha)}\right)^{(z)} \mathbf{R}_\alpha + \eta_{\hexagon} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} (\mathbf{r}_l - \mathbf{r}_k),
\end{aligned} \tag{21}$$

where, remarkably, the integral in the second summand evaluated to the second moment of area $\eta_{\hexagon}$ of the (hexagonal) plaquette shape, with

$$\eta_{\hexagon} := \int_{A_{\hexagon}} \mathrm{d}^2 s\, s_x^2 = \int_{A_{\hexagon}} \mathrm{d}^2 s\, s_y^2 = \frac{5\sqrt{3}}{16} a_{\mathrm{NN}}^4, \tag{22}$$

for our case of regular hexagons [21]. This leads to a decomposition of the polarization in two sums:

$$\begin{aligned}
\hat{\mathbf{P}}_{\mathrm{orb}} &= \frac{1}{A_{\mathrm{tot}}} \sum_\alpha \int_{A_{\hexagon}} \mathrm{d}^2 s\, \hat{\ell}^{(z)}(\mathbf{R}_\alpha + \mathbf{s})\, (\mathbf{R}_\alpha + \mathbf{s}) \\
&= \frac{-m_e}{A_{\mathrm{tot}}} \sum_\alpha \sum_{(k,l) \in \hexagon_{\mathbf{R}_\alpha}} \left(\frac{\sqrt{3}}{2} \mathbf{R}_\alpha + \frac{5}{24} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} (\mathbf{r}_l - \mathbf{r}_k) \right) \hat{J}_{k \leftarrow l} \qquad (23) \\
&= \hat{\mathbf{P}}_{\mathrm{orb}}^{(1)} + \hat{\mathbf{P}}_{\mathrm{orb}}^{(2)}, \qquad (24)
\end{aligned}$$

with

$$\hat{\mathbf{P}}_{\mathrm{orb}}^{(1)} := -\frac{\sqrt{3}\, m_e}{2\, A_{\mathrm{tot}}} \sum_\alpha \mathbf{R}_\alpha \sum_{(k,l) \in \hexagon_{\mathbf{R}_\alpha}} \hat{J}_{k \leftarrow l} \tag{25}$$

describing a dipole moment of loop current OAM and

$$\hat{\mathbf{P}}_{\mathrm{orb}}^{(2)} := -\frac{5\, m_e}{24\, A_{\mathrm{tot}}} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \sum_\alpha \sum_{(k,l) \in \hexagon_{\mathbf{R}_\alpha}} (\mathbf{r}_l - \mathbf{r}_k) \hat{J}_{k \leftarrow l} \tag{26}$$

describing a local polarization that depends on the plaquette shape via the second moment of area but not on the coordinate origin.

Alternatively, we can recognize that each nearest neighbor bond is shared by two plaquettes unless it sits on the border of our system. For the summands belonging to $\hat{\mathbf{P}}_{\mathrm{orb}}^{(1)}$, the two terms have opposite signs due to the anti-clockwise ordering required for $\theta_{kl}^{(\alpha)}$ and $\hat{J}_{k \leftarrow l} = -\hat{J}_{l \leftarrow k}$. The summands belonging to the shape polarization $\hat{\mathbf{P}}_{\mathrm{orb}}^{(2)}$ appear twice with the same sign. Note that for two plaquettes $\mathbf{R}_\alpha, \mathbf{R}_\beta$ that have the common bond $\mathbf{r}_l - \mathbf{r}_k$ it holds that

$$\mathbf{R}_\beta - \mathbf{R}_\alpha = \sqrt{3} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} (\mathbf{r}_l - \mathbf{r}_k), \tag{27}$$

such that $\hat{\mathbf{P}}_{\mathrm{orb}} = \hat{\mathbf{P}}_{\mathrm{orb}}^{\mathrm{bulk}} + \hat{\mathbf{P}}_{\mathrm{orb}}^{\mathrm{surf}}$ splits in a bulk term

$$\hat{\mathbf{P}}_{\mathrm{orb}}^{\mathrm{bulk}} := -\frac{23}{12} \frac{m_e}{A_{\mathrm{tot}}} \sum_{\langle k,l \rangle} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} (\mathbf{r}_l - \mathbf{r}_k) \hat{J}_{k \leftarrow l}, \tag{28}$$

and a surface term

$$\hat{\mathbf{P}}_{\text{orb}}^{\text{surf}} := \frac{m_e}{A_{\text{tot}}} \sum_{\alpha \notin L} \sum_{(k,l) \in \circlearrowleft_{\mathbf{R}_\alpha} \cap \text{NN}(L)} \left(\frac{\sqrt{3}}{2} \mathbf{R}_\alpha + \frac{5}{24} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} (\mathbf{r}_l - \mathbf{r}_k) \right) \hat{J}_{k \leftarrow l} \tag{29}$$

with the two sums meaning "for all plaquettes α outside the lattice L, sum over all bonds in $\circlearrowleft_{\mathbf{R}_\alpha}$ that are also part of the lattice bonds $\text{NN}(L) := \{\langle k, l \rangle\}$ if there are any". The surface term is difficult to write down, but its role is to simply subtract all summands of eq. (23) that we counted twice in the bulk term. Hence the surface term only depends on bond currents at the boundary of the lattice. Similar to the argument behind Stokes' theorem, most of the bulk contributions cancel out while the surface term summands are proportional to vectors with lengths of the order of the system size.

Further, note that a shift of the coordinate origin by $\mathbf{a}$ shifts the orbital polarization $\hat{\mathbf{P}}_{\text{orb}}$ by an integral over the OAM operator with respect to its origin,

$$\Delta \hat{\mathbf{P}}_{\text{orb}} = \frac{\mathbf{a}}{A_{\text{tot}}} \int_{A_{\text{tot}}} \mathrm{d}^2 r \, \hat{\ell}^{(z)}(\mathbf{r}). \tag{30}$$

It seems plausible that this integral should vanish. For the simulated systems this was the case, as numerical verification showed.

4.5 Orbital Hall Conductivity

The OHE is quantified by the OHC σ_{OH}. Assuming a linear response regime, we expect

$$\hat{\mathbf{J}}_{\text{orb}}^{(z)}(t) = \operatorname{Re} \sigma_{\text{OH}} \, \mathbf{E}(t), \tag{31}$$

where we allow complex $\sigma_{\text{OH}} \in \mathbb{C}^{2\times 2}$ to account for phase shifts.

The orbital Hall current is expected (and observed in our simulations) to be perpendicular to the electric field, i.e., we are interested in the following projection:

$$\hat{J}_{\text{orb}}^{(x)}(t) = \sigma_{\text{OH}}^{(xy)} \, E^{(y)}(t). \tag{32}$$

5 Implementation

Now that we have defined the Hamiltonian and operators of our system, we briefly look at how to numerically obtain its time evolution. More details can be found in the Python code on GitHub [22].

5.1 Time Evolution Method

We identify the site basis vector $|i\rangle$ with the complex standard basis vector

$$|i\rangle = (\delta_{1i}, \delta_{2i}, \ldots, \delta_{Ni}) \in \mathbb{C}^N, \tag{33}$$

such that all operators, e.g., the Hamiltonian $\hat{\mathcal{H}}(t)$, are matrices with elements $\mathcal{H}_{ij} = \langle i | \hat{\mathcal{H}} | j \rangle$ that we have already discussed in section 3. Since only on-site and nearest neighbor elements of $\hat{\mathcal{H}}(t)$ are finite, we store it as a sparse matrix.

To prepare the inital state with $E_F = 0$, we obtain the system eigenstates ($|\psi_n\rangle$) and energy levels (E_n) as the eigenvectors and -values of the matrix $\hat{\mathcal{H}}(0) = \hat{\mathcal{H}}_0$. Figure 3 shows the spectrum and two eigenstates of $\hat{\mathcal{H}}_0$ for an example system. The many-body density matrix $\hat{\rho}_0$ of the zero-temperature ground state is

$$\hat{\rho}_0 = \sum_{E_n \leq 0} |\psi_n\rangle\langle\psi_n| \,. \tag{34}$$

For finite times t, we obtain the density matrix $\hat{\rho}(t)$ by integrating the von Neumann equation from $\hat{\rho}(0) = \hat{\rho}_0$,

$$\frac{\partial \hat{\rho}(t)}{\partial t} = -i\Big[\hat{\mathcal{H}}(t), \hat{\rho}(t)\Big] + \frac{1}{\tau}\,(\hat{\rho}_0 - \hat{\rho}(t)), \tag{35}$$

where we added a simple decay term with relaxation time τ [23]. Equation (35) is integrated using a Runge-Kutta 4th order method, which provides $\hat{\rho}(t)$ at discrete timesteps $t \in \Delta t\,\mathbb{N}$. To check the convergence of the algorithm, each simulation was repeated with doubled and quadrupled time step divisions $\frac{\Delta t}{2}, \frac{\Delta t}{4}$.[1]

Expectation values of observables $\hat{O}$ are obtained from the density matrix via the trace, $\Big\langle \hat{O} \Big\rangle = \mathrm{tr}\Big(\hat{\rho}\,\hat{O}\Big)$. Hence, with eq. (8),

$$\Big\langle \hat{J}_{k\leftarrow l} \Big\rangle = 2\, t_{\mathrm{hop}}\, \mathrm{Im}\, \rho_{kl}, \tag{36}$$

and with eq. (23), we obtain

$$\Big\langle \hat{\mathbf{P}}_{\mathrm{orb}} \Big\rangle = -\frac{m_e}{A_{\mathrm{tot}}} \sum_{\alpha} \sum_{(k,l)\in \circlearrowleft_{\mathbf{R}_\alpha}} \left(\sqrt{3}\,\mathbf{R}_\alpha + \frac{5}{12}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} (\mathbf{r}_l - \mathbf{r}_k) \right) \mathrm{Im}\, \rho_{kl}. \tag{37}$$

The average orbital current is approximated by a forward difference,

$$\Big\langle \hat{\mathbf{J}}^{(z)}_{\mathrm{orb}}(t) \Big\rangle \approx \frac{1}{\Delta t}\Big(\Big\langle \hat{\mathbf{P}}_{\mathrm{orb}}(t+\Delta t) \Big\rangle - \Big\langle \hat{\mathbf{P}}_{\mathrm{orb}}(t) \Big\rangle \Big). \tag{38}$$

To get rid of artifacts beyond the linear OHE response, we extract the OHC from the fourier coefficients. Fourier-transforming eq. (32) provides

$$\sigma^{(xy)}_{\mathrm{OH}} = \frac{\mathcal{F}_\omega \Big\langle \hat{J}^{(x)}_{\mathrm{orb}} \Big\rangle}{E_0\, \mathcal{F}_\omega\, \epsilon}, \tag{39}$$

with the discrete fourier transform

$$\mathcal{F}_\omega\, \epsilon := \sum_n \epsilon(n\Delta t)\, e^{-i\omega\, n\Delta t}. \tag{40}$$

[1] The root mean square deviation between the orbital currents from the Δt, $\frac{\Delta t}{2}$ runs and the $\frac{\Delta t}{4}$ run were compared. For test systems it was checked that their ratio corresponds to a converged error of 4th order. For the run over all systems, single floating point precision was used to reduce simulation time; the resulting deviations were all on the order of the machine epsilon [24].

5.2 Simulation Parameters

We use natural units with $1 = t_{\text{hop}} = a_{\text{NN}} = \hbar = e$. To obtain a value for the electron mass m_e, we plug the parameters established for monolayer graphene, $t_{\text{hop}} = 2.8\,\text{eV}, a_{\text{NN}} = 0.142\,\text{nm}$ [17]. From $1 = \hbar^2\, t_{\text{hop}}^{-1}\, a_{\text{NN}}^{-2} = 1.23 \times 10^{-30}\,\text{kg}$ we obtain $m_e = 0.741$.

Approximating the k_y-modes for fixed k_x and $k_y \in (0, \pi)$ (of which a respective amount of L_y exists [16]) as equidistant results in the mode spacing $\Delta k_y = \frac{\pi}{L_y+1}$. Then the level spacing ΔE_y at the Fermi level follows as

$$\Delta E_y \approx \Delta k_y \left.\frac{\partial E}{\partial k}\right|_{E_F=0} = v_F\, \Delta k_y = \frac{3}{2}\frac{\pi}{L_y+1}, \tag{41}$$

where we used the Fermi velocity $v_F = \left.\frac{\partial E}{\partial k}\right|_{E_F=0} = \frac{3}{2}$ of the infinite lattice [17]. Hence, to observe interesting but not chaotic dynamics, we want to choose the frequency of the driving electric field to satisfy

$$\Delta E_y < \omega < W, \tag{42}$$

where W is the bandwidth. If we estimate the frequency ω_c at which wave packets reflect between the boundaries parallel to the x-axis by $\omega_c \approx \frac{2\pi}{2((L_y+1)/v_F)} = v_F \frac{\pi}{L_y+1} = \Delta E_y$, we also arrive at $\omega > \omega_c = \Delta E_y$ when minimizing the influence of standing-wave cavity modes on the system response. With a similar argument, we choose the relaxation time τ such that

$$\tau \lesssim L_y/v_F = \frac{2}{3}L_y, \tag{43}$$

to make it likely for electron wavepackets to scatter before reflecting at the open boundaries. Additionally, we impose

$$E_0 \ll t_{\text{hop}} \tag{44}$$

for the electric field amplitude to ensure the linear response regime.

Satisfying above equations led to simulations of system sizes[2] $L_x \in \{11, 19, 29\}, L_y \in \{20, 40, 80\}$ and driving frequencies $\omega \in \{0.5, 1, 2\}$ to fulfill eq. (42) with $W = 6$ for tight-binding monolayer graphene [17]. The Fermi level is kept at $E_F = 0$, the relaxation time is set to $\tau = 14$ following eq. (43), the electric field amplitude is $E_0 = 1 \times 10^{-3}$ to obey eq. (44). The field is ramped up over a time of $T_{\text{ramp}} = 12\pi$ to minimize beating from transients.

After T_{ramp}, 17 periods were simulated, resulting in the total simulation time of $T_{\text{ramp}} + 17 \times \frac{2\pi}{\omega}$. σ_{OH} is calculated from the last 10 periods where all systems were found to have reached a steady-state oscillation.

6 Results

The obtained OHCs are listed in table 1. As far as one can tell with the limited data points, the OHCs of constant driving frequency ω seem to converge for larger systems. See fig. 4 for a plot of the OHCs against the system size $L_x \cdot L_y$.

For the OHC mechanism we suspected in section 2.1, OBC were only crucial in the x-direction, thus it is not surprising that the influence of L_y on the measured OHCs is

[2] Odd L_x and even L_y are needed for a well-defined number of plaquettes. The resulting symmetry ensures $\hat{\mathbf{P}}_{\text{orb}}$ is origin-independent, as mentioned in section 4.4.

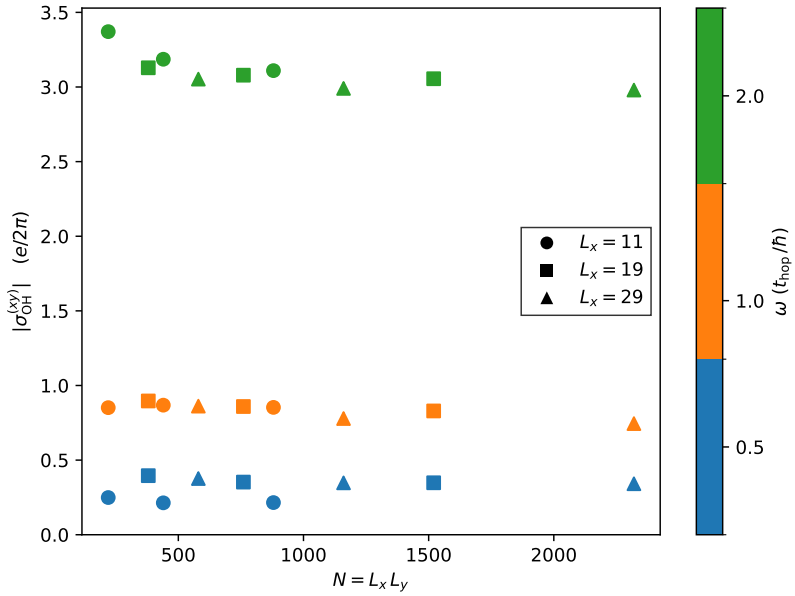

Figure 4. Magnitude of the obtained OHCs, plotted against the number of sites in the system $N = L_x \cdot L_y$. The color of the markers indicates the frequency ω of the driving field, the shape indicates the width L_x.

insignificant. It is interesting that σ_{OH} is also approximately constant for increasing L_x. However, since we used the polarization current and imposed the continuity equation (17), the influence of the OBC does not decrease for larger systems: if a boundary region of constant size contributes a certain amount of OAM to the orbital polarization, while ring currents (and thereby the local OAM) vanish in the bulk where Bloch's theorem is better fulfilled, the orbital polarization current is independent of the bulk size – we have assumed that the current has no sources, so it needs to traverse the entire crystal.

We obtain further insights on whether the OHE happens only at the boundaries from the real space currents. Figure 5 shows the bond currents and the resulting plaquette-wise OAM $\hat{\ell}(\mathbf{R}_\alpha)$ for the three different widths L_x and $L_y = 40, \omega = 1\,\frac{t_{\mathrm{hop}}}{\hbar}$ at the time were the orbital polarization $\hat{P}_{\mathrm{orb}}^{(x)}$ is maximal. Indeed, in all three cases, the plaquettes along the open armchair boundaries carry the most OAM due to stronger bond currents at the edges. However, apart from the boundaries, there are more streamlines of stronger currents, e.g. at the center of the $L_x = 11$ system in fig. 5 (a) or along dimer lines $6, 7$ and $13, 14$ for $L_x = 19$, fig. 5 (b). The touching plaquettes also have stronger OAM. Streamlines in AGNRs are expected [6] and would be a possibility to justify the constant OHC for larger L_x where orbital currents do not traverse the whole lattice and instead transport OAM between streamlines. A further investigation of the role of streamlines for an OHE may seem interesting but will be left to future research, for which it might be necessary to work with contributions beyond single plaquettes or a more complete theory such as in Ref. [13].

Further, interesting dynamics happen in the high driving frequency case $\omega = 2\,\frac{t_{\mathrm{hop}}}{\hbar}$: the phase of $\sigma_{\mathrm{OH}}^{(xy)}$ jumps by π and its magnitude is significantly larger, indicating a resonance-like behavior. The value $\omega = 2\,\frac{t_{\mathrm{hop}}}{\hbar}$ fits the two van Hove singularities in graphene at $\pm t_{\mathrm{hop}}$ (note the respective plateaus of states in the spectrum of fig. 3) such that indeed many

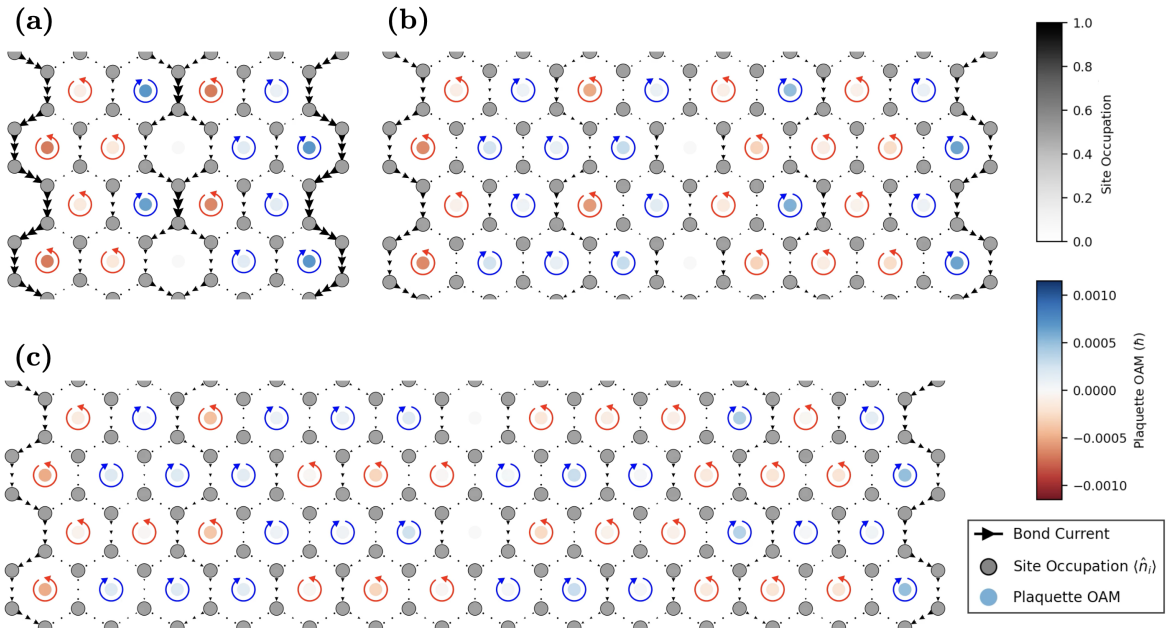

Figure 5. Parts of systems with driving field frequency $\omega = 1\,\frac{t_{\text{hop}}}{\hbar}$, with $L_y = 40$ and different widths: (a) $L_x = 11$, (b) $L_x = 19$, (c) $L_x = 29$. Systems are shown at the time when the orbital polarization $\hat{P}^{(x)}_{\text{orb}}$ is maximal (later in the simulation when the system is in steady-state oscillation). The center parts of the lattice, i.e., sites with $y \approx L_y/2$ are shown. The grey circles are the sites, their darkness indicates the on-site density $\langle \hat{n}_i \rangle$. The shade of the red or blue circles at the plaquette center indicates the total OAM $\int_{A_{\circlearrowleft}} \mathrm{d}^2 s\, \hat{\ell}^{(z)}(\mathbf{R}_\alpha + \mathbf{s}) = A_{\circlearrowleft}\, \hat{\ell}^{(z)}(\mathbf{R}_\alpha)$ of the plaquette. The black arrows between the sites are bond currents, the arrow size is proportional to the magnitude of the current. Note that the OAM is strongest at the armchair boundaries and also near current streamlines in the bulk.

interband transitions are available. Also, at a third of the bandwidth, ω is far out of the intraband dynamics regime.

We can see the frequency effects in the real space current patterns in fig. 6: for $\omega = 0.5, 1\,\frac{t_{\text{hop}}}{\hbar}$, the currents exhibit the known streamline patterns that entail plaquettes with larger OAM. Note hereby that the streamlines at $\omega = 0.5\,\frac{t_{\text{hop}}}{\hbar}$ seem to be packed more densely across the x-direction, which would make sense from the orbital current perspective: the currents between streamlines travel less distance, hence the OHC is lower. For $\omega = 2\,\frac{t_{\text{hop}}}{\hbar}$, the picture looks completely different, instead of streamlines, every second row of dimer lines has large uniform vertical currents. At the boundaries, the magnitude of the OAM is still larger, but with opposite cause: the currents are weaker along the armchair edge. This is also how the opposite phase of the OHC manifests. Within the bulk, the uniformity of the vertical currents leaves little OAM on the plaquettes.[3]

Finally, if we just look at the order of our obtained OHCs, we find that they are similar to comparable literature values. E.g., $\sigma^{(xy)}_{\text{OH}} \approx 1.5\,\frac{e}{2\pi}$ was found for a 512×512 flake of gapped graphene in Ref. [13] (DC case, using a Kubo formula). For a monolayer transition metal

[3]Note that in fig. 6, the orbital polarization of the lattice in the bulk seems perfectly split in a positive and negative half, but this is not the case for $L_x = 29$. Also, if we had looked at the point of maximum orbital polarization instead of maximum field amplitude, the OAM would be concentrated even stronger towards the edge plaquettes. Animations of all systems of a full period are available in the GitHub repository [22].

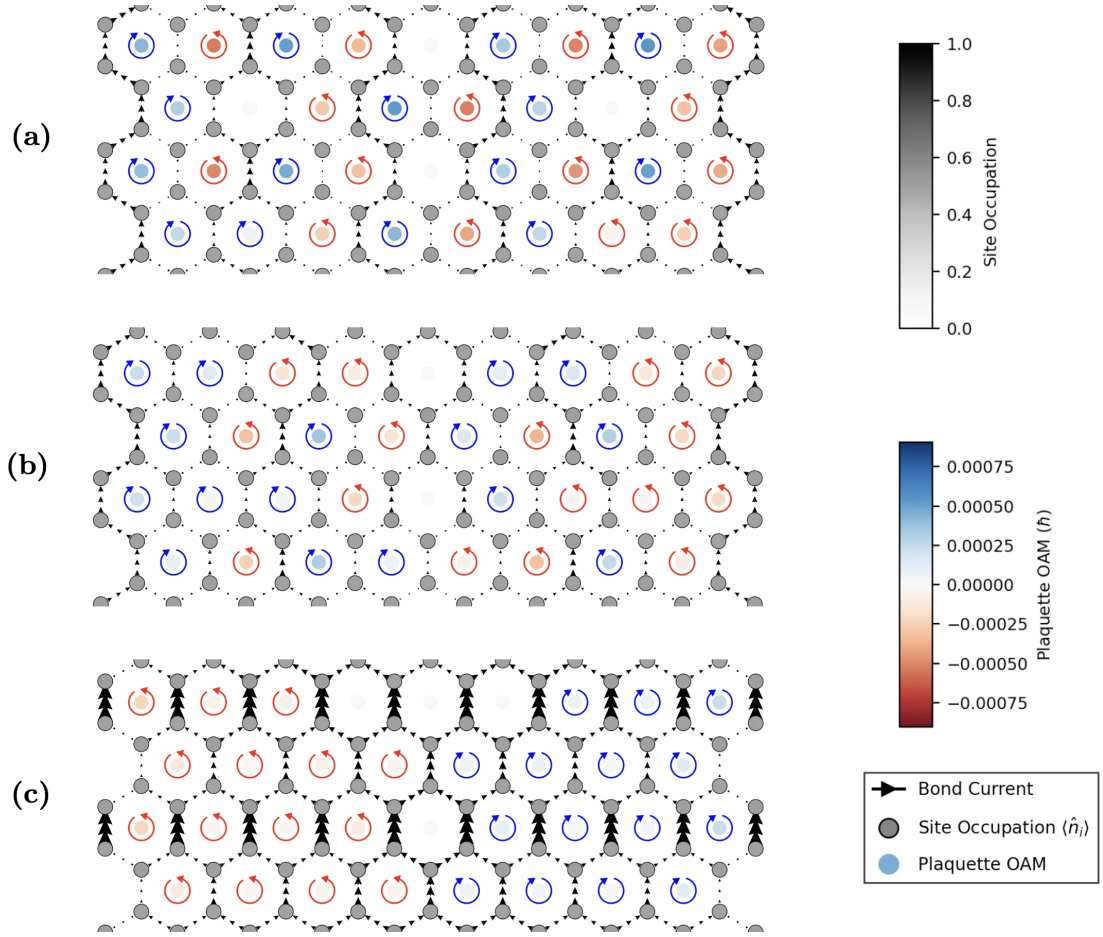

Figure 6. Parts of systems with different driving field frequencies: (a) $\omega = 0.5\,\frac{t_{\text{hop}}}{\hbar}$, (b) $\omega = 1\,\frac{t_{\text{hop}}}{\hbar}$, (c) $\omega = 2\,\frac{t_{\text{hop}}}{\hbar}$. All systems have dimensions $L_x = 19, L_y = 40$. Systems are shown at the time when the driving field amplitude is maximal, $\epsilon(t) = E_0$ (later in the simulation when the system is in steady-state oscillation). Zigzag rows around $y \approx \frac{3}{4}L_y$ are shown since the currents were the strongest there. The plots have the same structure as in fig. 5. Note the dramatic shift to $\omega = 2$ where vertical currents are uniformly strong and the orbital polarization is flipped.

dichalcogenide, Ref. [25] found AC OHCs of 1–2 $\frac{e}{2\pi}$ with peaks at $\hbar\omega \approx 1\,\text{eV}$ where the OHC is on the order of $\sigma_{\text{OH}}^{(xy)} \approx 10\,\frac{e}{2\pi}$. Ref. [26] managed to tune the AC OHC in gapped bilayer graphene up to triple digit multiples of $\frac{e}{2\pi}$ and found a sign change in the real part of the OHC at frequencies $\hbar\omega$ on the band gap order. Note that a direct comparison is difficult due to our different method with remaining corrections to the definition of $\hat{\ell}^{(z)}$ and source terms in the continuity equation. Still, these values render it very likely that an OBC-induced intersite current OHE in graphene is relevant.

7 Conclusion and Outlook

Increasing the range of materials for applications compared to spintronics, which depends on spin orbit interaction, is a central motivation for orbitronics [3]. Therefore it is useful to study how OBC can be used to generate an OHE in otherwise unsuited materials. For this, we introduced real space tight-binding operators based on the intuitive idea of

ring current OAM to investigate the OHE in finite graphene-like systems from intersite currents.

We found that the magnitudes of the simulated OHCs are similar to those of periodic systems. This strongly suggests that open boundaries induce a relevant AC OHE in graphene. At lower frequencies, the OHE results from current streamlines, primarily along the open boundaries, while the high frequency case exhibits uniform currents that result in a resonant, opposite-phase response of the orbital polarization with much larger OHC.

To strengthen these results, further considerations on our method could be taken. Primarily, the needed plaquette-wise truncation and source-free continuity equation are assumptions with unknown accuracy. Different real space methods [13, 14] could be used to verify the approach. Also, the decomposition of the orbital polarization into bulk and surface terms raises the question whether the bulk observable $\hat{\mathbf{P}}^{\text{bulk}}_{\text{orb}}$ (28) is more appropriate, as it eliminates any origin dependence and takes the OAM from neighboring plaquettes outside the lattice into account.

There are many other aspects that will be interesting to investigate further with our current method. Questions include whether the results transfer to wider systems, systems with periodic boundaries in the y-direction, different lattice structures or systems with intraatomic OAM contributions. Also, one could look for other phenomena of orbitronics such as the orbital Edelstein effect [27].

Table 1. Magnitude and phase of the OHC for different system sizes (L_x, L_y) and driving frequencies ω.

ω $(t_{\text{hop}}/\hbar)$	L_x	L_y	$\lvert\sigma_{\text{OH}}^{(xy)}\rvert$ $(e/2\pi)$	$\arg\sigma_{\text{OH}}^{(xy)}$ (rad)
0.5	11	20	0.249	−2.55
		40	0.214	−2.81
		80	0.215	−2.93
	19	20	0.396	−2.36
		40	0.353	−2.37
		80	0.348	−2.44
	29	20	0.377	−2.22
		40	0.348	−2.29
		80	0.342	−2.34
1.0	11	20	0.851	−2.37
		40	0.868	−2.42
		80	0.853	−2.42
	19	20	0.896	−2.66
		40	0.859	−2.73
		80	0.829	−2.77
	29	20	0.862	−2.62
		40	0.779	−2.67
		80	0.745	−2.71
2.0	11	20	3.371	1.60
		40	3.186	1.49
		80	3.110	1.46
	19	20	3.129	1.67
		40	3.079	1.66
		80	3.055	1.63
	29	20	3.053	1.79
		40	2.991	1.73
		80	2.980	1.70

Acronyms

ACA atom-centered approximation

AGNR armchair graphene nanoribbon

OAM orbital angular momentum

OBC open boundary conditions

OHC orbital Hall conductivity

OHE orbital Hall effect

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Schriftliche Versicherung

Ich habe die vorliegende Arbeit selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt. Über wissenschaftlich korrektes Arbeiten und Zitieren wurde ich aufgeklärt, und ich habe Kenntnis von möglichen Rechtsfolgen gemäß §24 Abs. 6 der Prüfungs- und Studienordnung für den Bachelor- und Masterstudiengang Physik an der Universität Regensburg. Ich versichere, dass die vorgelegten Druckexemplare und die vorgelegte elektronische Version der Arbeit identisch sind, und dass die Arbeit bisher keiner anderen Prüfungsbehörde vorgelegt wurde.